

Generation Interconnection System Impact Study Report

For Queue Project Q17-0905

Dresden Generating Station 230 kV

November 19, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0905. The Interconnection Customer proposes a 174.4 MW natural gas combined cycle facility interconnecting at the Dresden Generating Station 230 kV substation in the Illinois area. The request was received on December 16, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 270 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,946,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Farmersville Substation-Cuba City 100.0 kV Line

Contingency: Jones Chapel-Dekalb 500.0 kV

Loading: 116.5% (Rating: 322 MVA)

DFAX: 5.3%

Pre-project loading: 113.6%

2. Facility: Robison Park-Russell Fork 230.0 kV

Contingency: Dekalb-Jones Chapel 500.0 kV

Loading: 116.4% (Rating: 2,461 MVA)

DFAX: 43.6%

Pre-project loading: 113.4%

3. Facility: Hebo-Barrett 230.0 kV

Contingency: Winthrop-Hebo 230.0 kV

Loading: 115.2% (Rating: 702 MVA)

DFAX: 5.7%

Pre-project loading: 113.8%

4. Facility: Laotto-Golden Gate 100.0 kV Line

Contingency: Dekalb-Jones Chapel 500.0 kV

Loading: 113.8% (Rating: 282 MVA)

DFAX: 5.3%

Pre-project loading: 110.5%

5. Facility: Kendallville-Rex Brown 100.0 kV

Contingency: Jones Chapel-Dekalb 500.0 kV

Loading: 113.3% (Rating: 361 MVA)

DFAX: 5.3%

Pre-project loading: 110.7%

6. Facility: Rex Brown-Montibello 100.0 kV L

Contingency: Dekalb-Jones Chapel 500.0 kV line

Loading: 112.0% (Rating: 380 MVA)

DFAX: 5.3%

Pre-project loading: 109.6%

7. Facility: Hebo-Winthrop 230.0 kV

Contingency: Hebo-Barrett 230.0 kV Line

Loading: 111.1% (Rating: 760 MVA)

DFAX: 5.5%

Pre-project loading: 109.9%

8. Facility: Ligonier-Montibello 100.0 kV

Contingency: Jones Chapel-Dekalb Substation 500.0 kV line

Loading: 111.1% (Rating: 394 MVA)

DFAX: 5.3%

Pre-project loading: 108.8%

9. Facility: Napoleon Peaking Station-Richland 230.0 kV

Contingency: Troy Energy Llc-Richland 500.0 kV line

Loading: 110.9% (Rating: 1,063 MVA)

DFAX: 13.1%

Pre-project loading: 108.8%

10. Facility: Segs 5-Hillcrest 100.0 kV

Contingency: Richland-Dekalb 500.0 kV L

Loading: 110.9% (Rating: 554 MVA)

DFAX: 7.2%

Pre-project loading: 108.6%

11. Facility: Pueblo Xtra 500-230 kV Transformer Bank

Contingency: Pueblo Xtra-Boca Raton 500.0 kV

Loading: 109.5% (Rating: 1,916 MVA)

DFAX: 40.9%

Pre-project loading: 105.8%

12. Facility: Larson St-Norwich 230.0 kV line

Contingency: Barrett-Norwich 500.0 kV L

Loading: 109.2% (Rating: 1,299 MVA)

DFAX: 8.8%

Pre-project loading: 108.0%

13. Facility: Highgate-Herald-Twin Cities 230.0 kV line

Contingency: Mason St 500/230 kV Auto

Loading: 107.5% (Rating: 1,098 MVA)

DFAX: 7.0%

Pre-project loading: 106.4%

14. Facility: Canal Rd-Newport Tie 230.0 kV

Contingency: Beaver-Trego 500.0 kV L

Loading: 107.0% (Rating: 1,173 MVA)

DFAX: 10.4%

Pre-project loading: 105.5%

15. Facility: Beaver-Eufaula 500.0 kV

Contingency: Trego Substation-Galion Generating Sta 500.0 kV L

Loading: 106.9% (Rating: 4,084 MVA)

DFAX: 29.8%

Pre-project loading: 105.6%

16. Facility: Dixon Landing-Gould 230.0 kV L

Contingency: Richland Substation-Troy Energy Llc 500.0 kV

Loading: 105.7% (Rating: 540 MVA)

DFAX: 8.6%

Pre-project loading: 102.9%

17. Facility: Gruver-Galeville 230.0 kV Line

Contingency: Eufaula Substation-Imperial 500.0 kV

Loading: 105.3% (Rating: 622 MVA)

DFAX: 6.6%

Pre-project loading: 103.5%

18. Facility: Dixon Landing-Troy Energy Llc 230.0 kV

Contingency: Troy Energy Llc-Richland 500.0 kV L

Loading: 105.1% (Rating: 558 MVA)

DFAX: 8.6%

Pre-project loading: 102.5%

19. Facility: Trego Substation 230 kV/1 kV XFMR

Contingency: Troy Energy Llc-Trego 500.0 kV

Loading: 104.8% (Rating: 829 MVA)

DFAX: 12.9%

Pre-project loading: 102.1%

20. Facility: Darlington Rd Substation-Wayne Trace 100.0 kV

Contingency: Richland-Dekalb 500.0 kV L

Loading: 104.8% (Rating: 461 MVA)

DFAX: 6.0%

Pre-project loading: 102.5%

21. Facility: E Methuen-Lonetree 500.0 kV

Contingency: Muskogee-Montpelier Electric Generating Station 500.0 kV Line

Loading: 104.5% (Rating: 3,188 MVA)

DFAX: 9.2%

Pre-project loading: 104.0%

22. Facility: Trego-Galion Generating Sta 500.0 kV line

Contingency: Trego-Beaver 500.0 kV

Loading: 104.3% (Rating: 2,687 MVA)

DFAX: 20.6%

Pre-project loading: 103.0%

23. Facility: Madeline-East Methuen 230.0 kV

Contingency: Gray'S Flat-E Methuen 230.0 kV

Loading: 104.2% (Rating: 1,333 MVA)

DFAX: 9.5%

Pre-project loading: 103.0%

24. Facility: Highgate-Trego 230.0 kV L

Contingency: Trego-Troy Energy Llc 500.0 kV

Loading: 103.7% (Rating: 652 MVA)

DFAX: 10.9%

Pre-project loading: 100.8%

25. Facility: Muskogee Substation-Montpelier Electric Generating Sta Substation 500.0 kV line

Contingency: Dekalb Substation-Richland 500.0 kV

Loading: 103.7% (Rating: 3,404 MVA)

DFAX: 20.4%

Pre-project loading: 102.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cub City-Farmersville 100.0 kV line	Rebuild line with higher-capacity conductor	\$6,540,000	2% (\$100,000)
Robison Park-Russell Fork 230.0 kV	Rebuild line with higher-capacity conductor	\$50,389,000	2% (\$773,000)
Barrett-Hebo 230.0 kV	Rebuild line with higher-capacity conductor	\$8,079,000	2% (\$124,000)
Laotto-Golden Gate 100.0 kV Line	Rebuild line with higher-capacity conductor	\$6,146,000	2% (\$94,000)
Kendallville-Rex Brown 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,649,000	2% (\$56,000)
Rex Brown-Montibello 100.0 kV L	Rebuild line with higher-capacity conductor	\$7,193,000	2% (\$110,000)
Hebo-Winthrop 230.0 kV	Rebuild line with higher-capacity conductor	\$10,202,000	2% (\$157,000)
Montibello-Ligonier Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$5,842,000	2% (\$89,000)
Napoleon Peaking Station-Richland 230.0 kV	Rebuild line with higher-capacity conductor	\$29,830,000	2% (\$461,000)
Segs 5-Hillcrest 100.0 kV	Rebuild line with higher-capacity conductor	\$2,691,000	2% (\$41,000)
Pueblo Xtra 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$27,999,000	2% (\$440,000)
Norwich-Larson St 230.0 kV	Reconductor line	\$3,953,000	2% (\$61,000)
Highgate-Herald-TwinCities 230.0 kV L	Reconductor line	\$8,873,000	2% (\$137,000)
Canal Rd-Newport Tie 230.0 kV	Reconductor line	\$9,533,000	2% (\$147,000)
Beaver-Eufaula 500.0 kV	Reconductor line	\$24,572,000	2% (\$379,000)
Dixon Landing-Gould Substation 230.0 kV line	Reconductor line	\$3,070,000	2% (\$48,000)
Gruver-Galeville 230.0 kV	Reconductor line	\$5,946,000	2% (\$92,000)
Dixon Landing-Troy Energy Llc 230.0 kV	Reconductor line	\$7,480,000	2% (\$116,000)
Trego Substation 230 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$8,885,000	2% (\$138,000)
Darlington Rd Substation-Wayne Trace 100.0 kV	Reconductor line	\$711,000	2% (\$11,000)
East Methuen-Lonetree 500.0 kV	Reconductor line	\$38,891,000	2% (\$594,000)
Trego-Galion Generating Sta 500.0 kV line	Reconductor line	\$59,350,000	2% (\$917,000)
Madeline-East Methuen 230.0 kV line	Reconductor line	\$14,735,000	2% (\$227,000)
Trego Substation-Highgate Substation 230.0 kV	Reconductor line	\$14,571,000	2% (\$226,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muskogee-Montpelier Electric Generating Station 500.0 kV Line	Reconductor line	\$91,570,000	2% (\$1,408,000)
	Total allocated to project		\$6,946,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.